

Stellar PSFs Misassign Light in Little Red Dots

Broadband point-spread functions from field stars can fake extended hosts in compact high-redshift sources.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for little red dots.

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Background

Little red dots are compact early-Universe sources whose host galaxies are inferred by subtracting stellar point-spread functions. Broadband color mismatches between source and stars can corrupt this subtraction.

It was unclear how much of the apparent extended emission in LRDs is real versus a PSF mismatch artifact.

What this paper is selling

The authors derive the physical and mathematical origin of this error and provide a correction. Applied to LRDs, it shows that a published 11.8% F444W host component is likely spurious.

Key results to remember

- A pure point source can produce wavelength-dependent residuals mimicking a Sérsic host.
- The spurious F444W allocation survives every tested nuclear spectrum, stellar reference, and detector position.
- The error is large enough to alter inferred sizes, luminosities, and masses of compact distant sources.

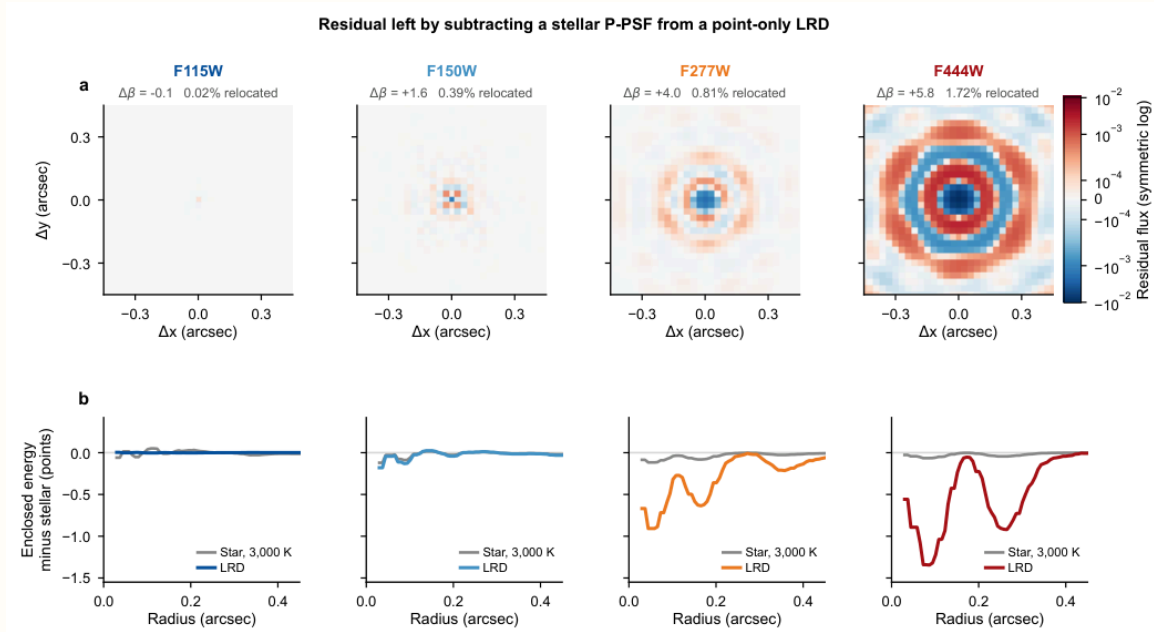


Figure 1 | Subtracting a stellar PSF from a point-only LRD leaves wavelength-dependent residuals that mimic a host galaxy.

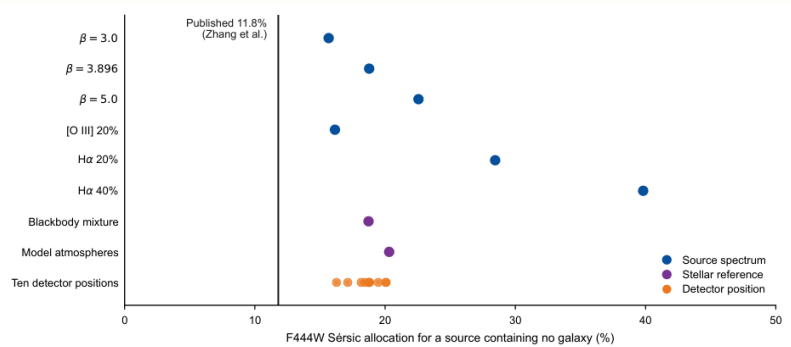


Figure 2 | Every tested LRD spectrum, stellar reference, and detector position yields a spurious F444W host above the published value.